ASHISH **KUMAR**

Ph.D. Candidate in Economics *(Johns Hopkins University)*

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Ashish Kumar

Skills

Statistics: causal inference, macroeconometrics, MonteCarlo simulation, bootstrapping, bagging, Bayesian inference, GMM

Machine Learning: dynamic machine learning techniques for forecasting and assessing uncertainty and risks in macroeconomics and finance

Mathematics: numerical optimization, dynamic programming

Economics: heterogeneous-agent lifecycle modeling, DSGE modeling, Bayesian VARs, Forecasting risk, HANK

Computing: parallelization, multithreading, probabilistic programming

Toolbox

Github Unix Latex

SQL MS Office CSS HTML

Docker AWS Redshift

Coding

Julia: Optim, DataFrames, Turing, Plots, Flux

Python: numpy, scipy, pandas, sklearn, XGBoost, LightGBM, pytorch, seaborn

R: tidyverse, dplyr, data.table, ggplot2

Misc: Fortran

Other

Languages: Hindi (native), English (fluent), Punjabi, French (beginner)

Education

2022 - now **Ph.D. Economics** — **Johns Hopkins University**, U.S.A.
Specialization: Macroeconomics with focus on heterogenous-agent lifecycle models
Advisors: *Christopher Carroll, Michael Keane*.

2018- 2020 **M.A. Economics**, magna cum laude — **Ashoka University**, India

2014- 2017 **B.Sc. Economics** — **Punjabi University**, India

Work Experience

2021 - 2022 **Economic Consultant** — **IHOPE Research Center**
External consultant for large interdisciplinary research center on health economics and public health. Provided econometric analysis and causal inference of proprietary data using R, SQL.

2020-2021 **Research Associate** — **Indian Institute of Management-Ahmedabad**
Provided economic and statistical analysis of pharmaceutical data for various research projects. Collaborated with large and diverse team and presented results to multiple audiences.

Research

- **Pension Design and Aggregate Savings: A Quantitative Assessment *(Work in Progress)***: I study how the shift from defined-benefit to defined-contribution pension system in the U.S. is affecting wealth inequality. DC balances are inheritable and depend on contributions and investment returns, while DB payouts reflect earnings history and end at death. I build a quantitative overlapping-generations model with rich household heterogeneity to see how households save in two different systems, and the resulting equilibrium outcomes. The model will be used to assess policy options—such as hybrid plan features and inheritance tax design—that can reduce inequality while staying fiscally sound.
- **Household Heterogeneity and Global Imbalances (Work in Progress)** *(with Julien Acalin (IMF))*
  We build a two-country heterogeneous-agent model that links welfare system, credit constraints, and growth beliefs to household savings and the current account imbalances. The framework separates structural vs belief-driven drivers of imbalances and tests reforms—entitlements, credit rules, and saving incentives—to inform IMF External Sector Assessments.
- **Optimal Taxation using Dynamic Stochastic Lifecycle Model**
  Developed a quantitative overlapping-generations model to evaluate different tax reforms, and using Monte-Carlo simulation method inferred that replacing capital-income tax with a flat wealth tax improve consumption, capital accumulation, and welfare while inheritance taxes produce mixed outcomes, thereby informing optimal tax policy trade-offs. Introduced new computational tools for the analysis of large dynamic stochastic lifecycle model with uncertainty.
- **Demographic Change and Automation: Implications for Wage and Wealth Inequality (Work in Progress)** *(with Kyung Woong Koh (JHU))*
- **Dynamic Life-Cycle Model Replication in Julia (Denardi 2004)**
  Replicated **Denardi (2004)** by developing a robust Julia implementation that solves complex dynamic life-cycle model of consumption-saving with voluntary and accidental bequests —code available on GitHub.